

INCIDENT REPORT

Network Attack Detection & Reporting - Simulated Honeypot Attack

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EXECUTIVE SUMMARY

A Cowrie SSH honeypot was deployed on an isolated VMware lab network and attacked from a Kali Linux VM to validate detection and logging capability. The exercise combined port scanning, manual credential attempts, and automated brute-forcing. All activity was captured via Cowrie's structured logs and a full Wireshark packet capture, confirming the honeypot successfully recorded attacker behavior end-to-end.

LAB ENVIRONMENT

- Attacker: Kali Linux VM (192.168.72.11)
- Target: Ubuntu Server VM running Cowrie honeypot (192.168.72.10)
- Network: Isolated VMware host-only segment "honeypot-isolated" (192.168.72.0/24), no internet access

ATTACK TIMELINE

Time (UTC)	Activity	Result
08:18 – 08:25	Nmap reconnaissance (TCP connect, service/version, OS fingerprint, script scan, full port scan) against 192.168.72.10	Identified port 22 (real OpenSSH, hardened) and port 2222 (Cowrie honeypot, spoofed Debian OpenSSH banner)
08:30:27	Manual SSH login attempt: root / admin123	Login accepted by honeypot
08:30:55	Manual SSH login attempt: root / toor commands: ls, exit	Login accepted; session and commands logged
08:31:30	Manual SSH login attempt: admin / 12345	Login accepted by honeypot
08:31:48	Manual SSH login attempt: test / 12345 commands: ls, exit	Login accepted; session and commands logged
08:33:33	Hydra automated brute-force (4 parallel threads) against root account	3 credential pairs (123456789, 12345, password) accepted within the same second, signature of automated tooling

KEY FINDINGS

- The real OpenSSH service on port 22 correctly rejected unauthorized root login attempts ("Permission denied"), demonstrating effective hardening on that service.
- The Cowrie honeypot on port 2222 accepted every credential pair submitted, by design, in order to bait and log attacker behavior rather than grant genuine system access.
- Cowrie's JSON log captured source IPs, ports, usernames, passwords, SSH client fingerprints (HASSH), typed commands, and full TTY session recordings for every session.
- Wireshark packet capture (honeypot_capture.pcap) corroborates the TCP/SSH handshake traffic between 192.168.72.11 and 192.168.72.10:2222.

INDICATORS OF COMPROMISE

Indicator	Value
Attacker (source) IP	192.168.72.11 (Kali Linux)
Target (honeypot) IP	192.168.72.10 (Cowrie sensor: ubuntuserver)
Target port	2222/tcp (SSH service emulation)
Credentials attempted	root/admin123, root/toor, admin/12345, test/12345, root/123456789, root/12345, root/password
Client SSH signatures	OpenSSH_10.3p1 Debian-4 (manual attempts), libssh_0.12.0 (Hydra automated attempts)

DEFENSE RECOMMENDATIONS

- Disable password-based root SSH login and enforce key-based authentication only, as already configured on this host's production SSH service.
- Deploy rate-limiting or account lockout (e.g., fail2ban) to blunt brute-force attempts.
- Monitor for burst login patterns: multiple successful/failed logins within sub-second intervals is a strong indicator of automated tooling.
- Operate honeypots on isolated network segments to safely gather threat intelligence without exposing production assets.
- Forward honeypot and authentication logs to a central SIEM for real-time alerting on brute-force and credential-stuffing patterns.

CONCLUSION

This exercise confirmed the honeypot's ability to detect, log, and provide actionable evidence of reconnaissance and credential-based attacks, while highlighting the contrast between a hardened production SSH service and a deliberately permissive decoy service.